VELSANET

White Paper No. 22

Velsanet H6-Based 6G RAN Architecture

Structure-Native Radio Access Without DU/CU

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Abstract

This white paper defines the architecture by which Velsanet's H6 node — the hexahedral autonomous node at the physical and RAN mapping tier — receives 6G ultra-small cell base stations through a dedicated face connection via MOCT ribbon cable. The document establishes that this structure eliminates the functional need for Distributed Units (DU) and Centralized Units (CU) as defined in O-RAN and 3GPP architectures, because the structural properties of H6 face connection and Velsanet optical-native E2E replace those functions at the architectural level.

The core argument is architectural, not incremental: in Velsanet, the topology precedes addressing, and the physical connection precedes protocol. When a 6G base station connects to an H6 face via ribbon cable, it becomes an entry point into the optical domain. From that moment, all network behavior is governed by Velsanet's Layer-1 optical architecture. No intermediate processing layer — DU or CU — is required or meaningful.

1. Purpose of This Document

This white paper extends Velsanet Wired–Wireless Convergence White Paper #02 by defining the precise physical and structural relationship between the H6 node and the 6G ultra-small cell base station.

Specifically, this document addresses:

- The physical installation model: base station on utility pole, H6 node below, connected by 5m+ MOCT ribbon cable
- The H6 face assignment model: one face dedicated to base station connection

- The structural reason DU and CU become architecturally unnecessary
- The wired–wireless convergence boundary as defined by the H6 face
- The relationship of this architecture to existing 6G / IMT-2030 standardization efforts

Scope Note

This document does not define wireless-to-optical channel mapping parameters. Those physical implementation details — modulation alignment, timing, power levels — are deferred to hardware testing after Velsanet Research Lab establishment. The structural architecture described here is complete and does not depend on those parameters.

2. The H6 Node — Structure and Role

2.1 H6 in the Velsanet Polyhedral Hierarchy

Velsanet defines five node types based on polyhedral geometry: T4 (tetrahedron, 4 faces), H6 (hexahedron/cube, 6 faces), O8 (octahedron, 8 faces), D12 (dodecahedron, 12 faces), and I20 (icosahedron, 20 faces). Each polyhedral type encodes structural role, connectivity capacity, and position in the network hierarchy.

H6 occupies the physical and RAN mapping tier. As defined in Velsanet White Paper #20 (Node Intelligence and Optical Parallelism):

> *H6 represents the physical and RAN mapping tier: the stable anchoring layer that bridges wireless and optical domains.*

H6 is an autonomous node. It possesses structural self-awareness — it knows its identity and its neighbors via Channel 1 of its ribbon cables — but it does not operate under AI layer coordination. H6 fulfills its role by existing and maintaining its physical connections. It does not route, prioritize, or respond to AI layer requests.

2.2 H6 Face Structure

The hexahedral geometry of H6 provides exactly 6 faces. Each face is a potential connection point. In the Velsanet architecture, face assignments follow structural role:

Face	Assignment	Description
Face 1	6G Base Station	Dedicated RAN connection face — MOCT ribbon cable from base station above
Face 2	O8 Upward	Connection toward O8 node — entry into Velsanet intelligent network layer
Face 3	Adjacent H6	Lateral connection to neighboring H6 node
Face 4	Adjacent H6	Lateral connection to neighboring H6 node
Face 5	Adjacent H6	Lateral connection to neighboring H6 node
Face 6	Reserved / Expansion	Available for additional base station or future structural extension

3. Physical Installation Model

3.1 Utility Pole Deployment

The H6-based 6G RAN deployment model is designed for ultra-small cell density in urban environments. The physical installation follows a two-point vertical model:

- 6G base station unit installed at the top of the utility pole — antenna exposure, wireless coverage
- H6 node installed at a lower point on the same pole, or at ground level / below-grade enclosure — structural connection point, optical domain entry
- MOCT ribbon cable (5 meters or more, factory pre-attached) runs vertically from the base station down to the H6 node face

```
[ 6G Ultra-Small Cell Base Station ]
     (Utility pole top — antenna)
            |
      Ribbon Cable
      (5m+, MOCT pre-attached,
       factory-fused, no connectors)
            |
            ↓
┌──────────────────────┐
│      H6 Node         │  ← Utility pole lower / ground enclosure
│    (C₆ Hexahedron)   │
│  [Face 1] ← Base Station (this face)
│  [Face 2] → O8 node (optical backhaul)
│  [Face 3] ↔ Adjacent H6
│  [Face 4] ↔ Adjacent H6
│  [Face 5] ↔ Adjacent H6
│  [Face 6]    Reserved
└──────────────────────┘
            |
     Velsanet Optical Network
            ↓
[ O8 Node — Intelligence begins here ]
```

3.2 MOCT Ribbon Cable Properties

The ribbon cable connecting the base station to the H6 face is a MOCT (Multi-Optical-Core Transceiver) ribbon cable as defined in Velsanet White Paper #03. Its key physical properties in this deployment context are:

- Factory pre-attached — fused at manufacture, shipped as an inseparable extension of the base station unit
- Minimum length 5 meters — sufficient to span the vertical distance from pole top to H6 node below
- No connectors — zero insertion-loss variation, zero contamination risk, no mechanical wear
- Fixed, non-branching — one ribbon, one H6 face; no patch panels, no splitters
- 8 physical channels per optical core — Channel 1 reserved for sensing, control, and structural identity bootstrap; Channels 2–8 for data

Physical Design Principle

The 5-meter ribbon cable is not a specification chosen for convenience. It reflects the MOCT architecture's elimination of connectors. In conventional optical systems, cable

length is constrained by connector placement. In Velsanet, the cable is the connection — it runs exactly as far as the physical gap between base station and H6 node requires, with no intermediate break points.

3.3 Urban Ultra-Small Cell Density

6G radio frequency bands (sub-THz, mmWave) require cell radii of tens of meters. This demands high-density base station deployment across urban environments. The H6 utility-pole model is structurally compatible with this density requirement:

- Every utility pole becomes a potential RAN node — no new civil infrastructure required
- H6 nodes connect laterally to adjacent H6 nodes via remaining faces — forming a ground-level mesh that mirrors the aerial antenna grid
- T4 nodes (personal/device-level) connect to the nearest H6 — preserving the rhizome origin principle of free, decentralized connection

4. Why DU and CU Are Not Required

4.1 What DU and CU Do in O-RAN

In the O-RAN and 3GPP architecture for 5G and proposed 6G, the radio access network is functionally split into three units:

- Radio Unit (RU) — antenna, RF processing, low-level PHY
- Distributed Unit (DU) — higher PHY, MAC, RLC — typically near the radio
- Centralized Unit (CU) — PDCP, RRC, session management — typically in the cloud or regional data center

This functional split exists because the underlying transport — IP/Ethernet — cannot guarantee deterministic end-to-end behavior. The DU and CU exist to manage packet scheduling, retransmission, session state, and routing across a network that does not provide structural guarantees. They are software abstractions compensating for physical-layer uncertainty.

4.2 What Replaces Them in Velsanet

When a 6G base station connects to an H6 face via MOCT ribbon cable, the following structural properties immediately apply:

- The connection is physically fixed — one ribbon, one face, zero ambiguity
- Channel 1 activates automatically — device authentication, physical alignment verification, structural identity exchange — without any external controller
- An individual E2E optical path is created — one device, one optical core, deterministic from base station to destination
- Handover is managed by V-MCU through overlapped E2E — the new E2E is pre-created before the old one is released, with zero packet loss
- All higher-layer network behavior is governed by Velsanet's structural topology — O8 nodes handle intelligence upward

Architectural Principle

DU and CU are not removed from Velsanet because they were made more efficient. They

are absent because the problems they solve do not exist in a topology-first, optical-native architecture. Packet scheduling is unnecessary when each device has a physically isolated optical core. Session management is structural when E2E paths are created and released by topology, not by software state machines.

4.3 Function-by-Function Comparison

Function	O-RAN DU / CU	Velsanet H6
Packet scheduling	DU — software scheduler	Not required — each device has isolated optical core
PHY/MAC processing	DU — higher PHY, MAC, RLC layers	H6 face + MOCT — structural optical mapping
Session management	CU — PDCP, RRC state machines	E2E path creation/release by topology (V-MCU)
Routing / forwarding	CU — IP routing, GTP tunneling	Velsanet E2E — no routing protocol required
Handover coordination	DU+CU — X2/Xn interface signaling	V-MCU — overlapped E2E, zero packet loss
Fronthaul / midhaul	eCPRI / F1 interface over Ethernet	MOCT ribbon cable — direct face connection
Backhaul	IP/MPLS transport network	Velsanet optical backbone via O8 upward

5. The Wired–Wireless Convergence Boundary

5.1 Wireless as Entry Point, Not Network

Velsanet Wired–Wireless Convergence White Paper #02 establishes the foundational principle:

> *Wireless access in Velsanet is not a network, but an entry point into the optical domain. The wireless signal is received at the radio unit. The signal is immediately mapped into an optical core. From that moment onward, the communication is handled entirely by Velsanet's Layer-1 optical architecture.*

The H6 face connection makes this principle physically concrete. The base station is the wireless domain. The H6 face is the convergence boundary. The MOCT ribbon cable is the crossing point. Everything that happens after the ribbon cable enters the H6 face is Velsanet — optical, structural, deterministic.

5.2 The Convergence Boundary Is Physical

In O-RAN, the convergence boundary between wireless and wired domains is a software interface — eCPRI, F1, N2/N3 — defined by protocol specifications that can be updated, versioned, and negotiated. The boundary is logical, not physical.

In Velsanet, the convergence boundary is the H6 face. It is a physical surface. The ribbon cable either connects to it or does not. There is no protocol negotiation at the boundary because the structural identity is established by Channel 1 of the ribbon cable before any data

flows. The boundary is not a handshake — it is a physical fact.

5.3 What the H6 Node Does Not Do

Understanding H6's role requires clarity about what it explicitly does not do. H6 is an autonomous node. It does not:

- Coordinate across multiple peers
- Prioritize optical paths
- Respond to requests from AI layers (PAI, AAI, AsAI)
- Perform packet processing of any kind
- Manage session state
- Implement any wireless protocol (LTE, NR, 6G NR)

H6 exists and maintains its connections. The base station sends wireless signals. The ribbon cable carries them into the H6 face. The H6 maps them into optical cores. That is the complete function.

6. Mobility Management — The O8-Centered System

6.1 Mobility as a System, Not a Node Function

Mobility management in Velsanet is not a function performed by a single component. It is a system behavior that emerges from the coordinated operation of four structural elements: the H6 node, the O8 node, the V-MCU, and the device Identity. Each element has a distinct and non-overlapping role. None of them alone manages mobility — the system does.

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Mobility Management System

┌──────────────────────────────────────────────────────┐
│                    O8 Node                           │
│  ┌──────────────────────────────────────────────┐    │
│  │  Identity Tracking   │  Signal Comparison    │    │
│  │  Overlap Decision    │  E2E Release Control  │    │
│  └──────────────────────────────────────────────┘    │
│                ↕ Commands ↕                          │
│             V-MCU (executor)                         │
│           ↙                ↘                         │
│   [H6-A]                [H6-B]                       │
│     ↑                      ↑                         │
│  Base Stn A            Base Stn B                    │
│     ↑                      ↑                         │
│            [Device — Identity fixed]                 │
└──────────────────────────────────────────────────────┘
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6.2 Why O8 Is the Mobility Decision Point

H6 is an autonomous node. It knows itself and its immediate neighbors. It cannot compare signal quality across multiple H6 nodes, cannot track a device across cells, and cannot decide when to pre-create or release an E2E path. H6 fulfills its role by existing — not by deciding.

O8 is the first node tier at which intelligence is structurally assigned. O8 can simultaneously observe multiple H6 nodes connected to it, compare their states, and issue coordinated commands through V-MCU. This is precisely the capability that mobility management requires.

Mobility decision authority therefore belongs to O8 — not because it was assigned by convention, but because the structural properties of O8 are the minimum required for mobility judgment.

6.3 Identity-Based Device Tracking

Every device that connects to Velsanet over a wireless interface carries a Velsanet Identity as defined in White Paper #15. The Identity is a network-native, structurally bound identifier — it does not change when the device moves, changes access point, or changes network layer.

When a device connects to a base station attached to H6-A, O8 records:

- Device Identity → H6-A face → optical core assignment
- Signal state via Channel 1 monitoring
- E2E path status (Active / Standby / Overlapping)

As the device moves, O8 tracks the Identity continuously. The Identity is the anchor. The H6 attachment point is the variable. Mobility is the management of that variable — without ever losing the Identity binding.

6.4 Overlapped E2E — The Mechanism of Zero-Loss Handover

The default handover mode in Velsanet is make-before-break, implemented through Overlapped E2E. The sequence is:

1	Device approaches H6-B coverage area. O8 detects signal presence on H6-B via Channel 1 monitoring.
2	O8 instructs V-MCU to pre-create a new E2E on H6-B for the same device Identity. H6-A E2E remains Active.
3	Both E2Es coexist — Overlapping state. O8 compares signal quality from both H6-A and H6-B simultaneously.
4	When H6-B signal is confirmed stable and H6-A signal is degrading, O8 instructs V-MCU to migrate traffic to H6-B E2E.
5	H6-B E2E becomes Active. H6-A E2E transitions to Terminating state and is released.
6	Identity binding is preserved throughout. No packet loss. No re-authentication. No session interruption.

6.5 Signal Selection — Choosing the Better Path

When a device is located between two H6 nodes — within coverage of both — O8 performs continuous signal comparison. This is not a handover event. It is ongoing signal selection: O8 observes Channel 1 data from both H6-A and H6-B and maintains the E2E on the structurally stronger path.

Signal selection operates as follows:

- Both H6-A and H6-B report Channel 1 status to O8 continuously
- O8 evaluates signal quality against predefined stability thresholds
- If H6-A is stronger: H6-A E2E remains Active, H6-B E2E held in Standby
- If H6-B becomes stronger and stable: O8 triggers make-before-break sequence
- The device experiences no disruption — the selection is network-side

This is structurally distinct from conventional handover: the device does not initiate, negotiate, or even detect the selection. The network selects on behalf of the Identity.

6.6 Role Boundaries Within the Mobility System

Component	Role in Mobility	Does NOT Do
H6 Node	Physical anchor — receives base station signal via ribbon cable face	Decision-making, signal comparison, E2E management
O8 Node	Mobility decision authority — Identity tracking, signal comparison, overlap control, E2E release command	Physical optical switching, direct device interaction
V-MCU	Execution layer — creates, activates, and releases E2E paths on O8 instruction	Deciding when to handover, comparing signal quality
Device Identity (WP15)	Persistent anchor — remains bound throughout movement, never re-authenticated on handover	Initiating or detecting handover

6.7 Mobility Across Multiple O8 Domains

When a device moves beyond the coverage area of a single O8 node — crossing into an adjacent O8 domain — mobility management escalates to the D12 layer (AAI). The O8-to-O8 transition follows the same Overlapped E2E principle, but coordination is handled by the D12 node above both O8 nodes.

This is consistent with the Velsanet hierarchical intelligence model: each layer manages mobility within its structural authority. O8 manages H6-to-H6 transitions. D12 manages O8-to-O8 transitions. The Identity remains bound throughout all levels of transition.

7. Comparison with 5G RAN Architecture

7.1 Structural Difference — One Principle

The fundamental difference between 5G and Velsanet H6-based 6G RAN is not a matter of speed, capacity, or efficiency. It is a matter of architectural class. 5G compensates for the limitations of packet-switched transport through software layers. Velsanet eliminates those limitations by replacing the transport model entirely.

> *5G is a system where software compensates for hardware limitations. Velsanet is a system where topology is the network function.*

7.2 Architecture Side by Side

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5G O-RAN:
Base Station (RU) --> DU --> CU --> Core Network
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[RF processing]     [PHY/MAC] [Session] [EPC/5GC]
Each step: software processing, protocol translation, latency

Velsanet H6:
Base Station --> Ribbon Cable --> H6 Face --> Optical Core --> Velsanet Network
[RF]            [5m MOCT]        [boundary]  [E2E path]      [O8 and above]
No intermediate processing. No protocol layers. No latency accumulation.
```

7.3 Item-by-Item Comparison

Dimension	5G O-RAN	Velsanet H6-Based RAN
RAN Structure	RU / DU / CU / Core — 3-stage functional split	Base station / H6 face / optical core — single boundary
Transport Model	Packet switching over eCPRI / IP / Ethernet	Physically isolated optical core per device
Fronthaul / Backhaul	eCPRI fronthaul, F1 midhaul, N3 backhaul — three interfaces	MOCT ribbon cable (5m+) — one physical connection, no interfaces
DU / CU	Required — PHY/MAC processing, session management, routing	Not required — H6 face and optical E2E replace these functions structurally
Handover Mechanism	X2/Xn signaling — DU reports, CU approves, device re-connects	Overlapped E2E — O8 pre-creates new path, transitions silently
Device Role in Handover	Device measures signal, reports to network, executes re-connection	Device does nothing — network manages Identity on device's behalf
Authentication on Move	Re-authentication possible on handover — session re-establishment	No re-authentication — Identity is persistent, bound at manufacturing
Mobility Decision	Device + DU + CU collaborative signaling	O8 node — sole authority based on Channel 1 signal comparison
AI Integration	External overlay — AI added above the network as service layer	Structural — PAI/AAI/AsAI embedded in O8/D12/I20 node tiers
Signal Selection	Device-initiated measurement reporting to network	O8 continuous Channel 1 monitoring — network-side, device-transparent
Latency Source	Packet queuing + DU/CU processing + re-routing on congestion	Physical path only — no queuing, no processing, no routing decisions
Security Model	Cryptographic protocols at upper layers — physical path unauthenticated	Physical-layer structural trust via Channel 1 at connection time

7.4 Why the Difference Is Architectural, Not Incremental

The 5G functional split — RU, DU, CU — exists because IP-based transport cannot provide deterministic end-to-end behavior. DU compensates with local scheduling. CU compensates with session state management. The O-RAN open interface standardizes the compensation.

None of this changes the underlying problem: packets compete, paths are logical, and reliability is probabilistic.

Velsanet does not improve upon this model. It replaces the premise. When each device has a physically isolated optical core from the H6 face to its destination, there is nothing to schedule, no session to manage, and no route to compute. The functions that DU and CU perform become structurally unnecessary — not because they were made faster, but because the conditions that created the need for them no longer exist.

This is the sense in which Velsanet and 5G O-RAN belong to different architectural classes. The primary distinction is that one architecture assumes packet-switched transport and builds compensating layers above it; the other assumes topology-first optical connectivity and finds no need for those layers at all.

8. Relationship to 6G and IMT-2030 Standardization

6.1 IMT-2030 Requirements and Velsanet

ITU-R M.2160 defines the framework and objectives for IMT-2030 (6G). Among its key requirements are sub-millisecond latency, extreme reliability, and AI-native network operation. The H6-based RAN architecture addresses these requirements structurally:

- Sub-millisecond latency — achieved by eliminating DU/CU processing layers and replacing packet-switched transport with physically isolated optical cores
- Extreme reliability — achieved by MOCT's connector-free, permanently fused ribbon cable architecture, eliminating the primary failure modes of conventional optical systems
- AI-native operation — achieved by Velsanet's three-layer AI structure (PAI/AAI/AsAI) embedded in O8 and above, operating on a topology that is structurally stable from the H6 layer upward

6.2 Position Relative to O-RAN

O-RAN Alliance has defined an open, disaggregated RAN architecture that separates RU, DU, and CU and standardizes interfaces between them. This is an evolutionary approach: it improves upon the closed, vendor-proprietary RAN of 4G/5G while retaining the fundamental IP/packet-switched transport model.

Velsanet's H6-based RAN is not an evolution of O-RAN. It is a structural alternative that replaces the transport model entirely. The RU–DU–CU functional split is an artifact of packet-switched transport requirements. When those requirements are eliminated by optical-native E2E, the split loses its justification.

This does not mean Velsanet is incompatible with 6G radio technology. The base station unit that connects to the H6 face can implement any 6G radio standard — sub-THz, mmWave, or any spectrum defined by ITU-R or 3GPP. The radio unit is external to Velsanet. Only the transport and network function begins at the H6 face.

6.3 Proposed Contribution to ITU-T SG13

This white paper provides architectural basis for the following addition to Velsanet's ITU-T SG13 contribution under Work Item B (Optical-Native E2E Connectivity):

> *The H6 hexahedral node, connected to a 6G radio unit via factory-fused MOCT ribbon cable, demonstrates that the RU–DU–CU functional split is not an inherent requirement of radio access network architecture. It is a consequence of packet-switched transport. In an optical-native, topology-first network, radio access is a face*

> *connection — structurally defined, physically permanent, and protocol-independent.*

9. Summary

This white paper establishes the following architectural principles:

1. The 6G ultra-small cell base station connects to one face of the H6 node via MOCT ribbon cable (5m+, factory pre-attached, no connectors).
2. H6 is installed below the base station — on the utility pole or at ground level. The ribbon cable spans the vertical distance between them.
3. The H6 face is the convergence boundary between the wireless domain and the Velsanet optical domain. It is a physical surface, not a software interface.
4. When the ribbon cable enters the H6 face, Channel 1 activates automatically — structural identity is established without external coordination.
5. From the H6 face inward, all network behavior is Velsanet — optical, structural, deterministic. No DU or CU layer exists or is required.
6. DU and CU are absent not because they were optimized away, but because the problems they solve — packet scheduling, session management, routing — do not arise in an optical-native, topology-first architecture.
7. Intelligence is introduced from O8 upward. H6 is autonomous — it fulfills its role by existing and maintaining its connections.
8. Mobility management is a system behavior: O8 is the decision authority, V-MCU is the executor, H6 is the physical anchor, and device Identity is the persistent binding throughout movement.
9. Overlapped E2E is the default handover mode — make-before-break, zero packet loss, no re-authentication, no session interruption.
10. Signal selection is network-side: O8 continuously compares Channel 1 quality from adjacent H6 nodes and selects the stronger path on behalf of the device Identity.
11. Wireless-to-optical channel mapping parameters are not defined in this document. The structural architecture is complete. Physical implementation values will be determined through hardware testing after Velsanet Research Lab establishment.

References

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